Avidan Kwok

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Education

University of British Columbia Expected Apr 2028
Bachelor of Commerce – Combined Major in Business and Computer Science (BUCS) *Vancouver, BC*
Relevant Coursework: Algorithms & Data Structures, Software Construction, Computer Systems, Models of Computation

Technical Projects

jobhunt | *Python, FastAPI, Claude API, PostgreSQL, Vercel, Oracle Cloud, pytest* Jul 2026 – Present
- Built a **$0-infra** Discord alert service for SWE internships, splitting a FastAPI signup app (Vercel) from a cron worker (Oracle Cloud) over Postgres
- Built 31 job-board integrations with schema-drift detection that surfaces scraper breakage instead of failing silently
- Designed a concurrent scheduler with per-host rate limits and a circuit breaker, scanning **403 companies in under 7 min**
- Paired employer-aware classification rules with a Claude Haiku tie-break for ambiguous titles, capped at $0.25 a day
- Encrypted webhooks at rest (Fernet) with HMAC dedup and SSRF-safe validation; 800+ tests across SQLite and Postgres

Altamyra | *TypeScript, React, Hono, Cloudflare Workers, Durable Objects, Gemini API* Nov 2026
- Led product and backend for an Individualized Education Program (IEP) platform for students, teachers, EAs, and parents in BC schools
- Architected a Cloudflare Workers backend with per-student Durable Objects, role-based access, and an audit log
- Designed a privacy-first Gemini pipeline that strips student names before every call and fails closed on any leak
- Built a Maritime agent that adapts lesson plans to IEP goals, with a **5-model** Gemini fallback chain
- Won **Third Place, Best Agent on Maritime** out of **76 teams** at nwPlus HackCamp 2026

Avi's Word Hunt Solver | *Python, Trie, DFS, wordfreq, JavaScript* Nov 2026
- Built a Word Hunt solver that finds every valid word on a 4x4 board in **3 ms** via lockstep trie traversal and DFS
- Pruned the search with a 16-bit visited bitmask and early trie cutoffs, exploring 2K of 12M possible paths per board
- **Cut cold start to 0.1 s** with a prebuilt dictionary cache; served a JSON API and web UI with 107 tests

Mushu AI | *Python, FastAPI, LangChain, Groq, ChromaDB, ONNX, React, Next.js* Jan 2026 – Mar 2026
- Designed and built a full-stack RAG chatbot (React, FastAPI, Groq) that answers as my dog Mushu Kwok from a curated corpus, with **p50 latency under five seconds**
- Cut deployment image size by **83% (5.7GB to under 1GB)** by replacing HuggingFace/torch with ONNX runtime
- Built LangChain ingestion into ChromaDB with ONNX embeddings, streaming Groq tokens to the client over SSE

Experience

TD Canada Trust May 2025 – May 2026
Customer Experience Associate *Vancouver, BC*
- Partnered with advisors on personalized client referrals, generating **$1.2M+** in investments and **18%** branch growth
- Earned a **100% customer score** in 2026 through clear, concise client service in English, Cantonese, and Mandarin

Awards

Third Place, Best Agent on Maritime (Sponsor Track) Nov 2026
nwPlus HackCamp 2026 *Vancouver, BC*

First Place, Product+ 2026 Oct 2026
UBC BizTech Product+ 2026 *Vancouver, BC*
- Won first place out of 29 teams and 150+ competitors in an SAP-sponsored product case
- Pitched and prototyped an AI agent that pre-positions pharmacy inventory ahead of flu-season demand surges

Technical Skills

Languages: Python, TypeScript, Java, C++, SQL
Frameworks: React, Next.js, FastAPI, Hono, LangChain
Cloud & Tools: Git, Linux, Cloudflare Workers, Oracle Cloud, pytest, Claude API, Gemini API